

Holistic Data Preparer

Theory Concepts Document Data Preprocessing and Feature Engineering

Project: Customer Credit Risk Prediction (Fintech)

This document explains all the theoretical concepts used in the project. It contains definitions, reasons, and intuition only — no code. The corresponding practical implementation is in the Jupyter Notebook file.

1. What is Data Analysis?

Data Analysis is the process of inspecting, cleaning, transforming and interpreting raw data in order to discover useful information, draw conclusions and support decision-making.

There are four main types of data analysis:

- Descriptive Analysis: tells us what happened (for example, average income, counts of defaults).
- Diagnostic Analysis: tells us why it happened (for example, why some customers default more).
- Predictive Analysis: tells us what might happen next (machine learning models).
- Prescriptive Analysis: suggests what action should be taken.

The typical steps are: define the question, collect data, clean data, explore the data, analyse or model, visualise results and communicate findings.

2. Planning a Data Science Project (CRISP-DM)

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a popular six-step framework:

1. Business Understanding — define the problem, objectives and success criteria.
2. Data Understanding — collect data, examine its types, shape, distributions and quality.
3. Data Preparation — clean, impute, encode, scale and engineer features (the main focus of this project).
4. Modeling — select and train machine learning models.
5. Evaluation — measure performance using suitable metrics.
6. Deployment — put the model into production and monitor it.

Good planning also includes choosing the target variable, selecting the right evaluation metric, handling time and resource constraints, and thinking about ethical issues.

3. Framing a Machine Learning Problem

To frame an ML problem we answer three questions:

7. What are the inputs (X)? — the features describing each customer.
8. What is the output (y)? — the target we want to predict.
9. What kind of problem is it?
 - If the output is a category, it is a Classification problem.
 - If the output is a number, it is a Regression problem.
 - If there are no labels, it is an Unsupervised / Clustering problem.

In this project the target is `default_flag` which can only be 0 (no default) or 1 (default), so it is a Binary Classification problem. Because defaulters are fewer than non-defaulters, the dataset is imbalanced. For this reason accuracy alone is misleading; metrics such as Precision, Recall, F1-score and ROC-AUC are more meaningful.

4. Tensors

A tensor is a multi-dimensional array of numbers. It is the basic data structure used in almost all machine learning and deep learning frameworks (NumPy, PyTorch, TensorFlow).

- Scalar (0-D tensor): a single number, for example 42.
- Vector (1-D tensor): a list of numbers, for example [10, 20, 30].
- Matrix (2-D tensor): a table with rows and columns, like a dataset.
- 3-D tensor: a collection of matrices, for example a batch of greyscale images.
- N-D tensor: higher-dimensional arrays, such as batches of RGB images or video.

A tabular dataset is therefore a 2-D tensor whose rows are samples and columns are features. Tensors support operations such as addition, multiplication, transposition and reshaping.

5. Data Acquisition from Multiple Sources

Real-world data rarely comes from one place. In this project we combine four sources:

- CSV (Comma-Separated Values): a plain-text table where each row is a record. It is simple, portable and widely used.
- JSON (JavaScript Object Notation): a key-value format, commonly returned by web APIs. Good for nested or semi-structured data.
- SQL Database: data stored in tables that can be queried with Structured Query Language. We use SQLite, a lightweight file-based database.

- External API: a web service that returns data. We use a dummy API that provides economic indicators and region-wise default rates.

After loading each source, we combine them into one master table using a common key, here `customer_id`. This is called data integration or merging.

6. Data Understanding and Exploration

Before cleaning, we must understand the data:

- `.info()` shows column names, data types and non-null counts.
- `.describe()` gives summary statistics (mean, standard deviation, min, max, quartiles) for numeric columns.
- `.nunique()` and `value_counts()` reveal how many distinct values exist and how often each appears.
- Pandas Profiling (ydata-profiling) generates an automated report containing missing values, distributions, histograms and correlations.

This exploration helps us decide how to clean and transform each column.

7. Missing Data and Imputation

Missing values are empty cells in the dataset. They can occur because of human error, system failures, privacy rules or customers not answering certain questions. Most ML models cannot handle missing values, so they must be handled.

Missingness types:

- Missing Completely at Random (MCAR): no pattern, e.g. a random system error.
- Missing at Random (MAR): missingness can be explained by other columns.
- Missing Not at Random (MNIR): the missingness itself carries information.

7.1 Complete Case Analysis

We simply drop rows (or columns) that contain missing values. It is fast but can remove a lot of data and introduce bias if missingness is not random.

7.2 Simple Imputer — Mean / Median

For numerical columns, missing values can be filled with the mean or the median. The median is preferred when the column has outliers, because it is not pulled towards extreme values.

7.3 Simple Imputer — Most Frequent (Mode)

For categorical columns, missing values are filled with the category that appears most often. It is simple and preserves the original distribution reasonably well.

7.4 Missing Indicator + Random Sample Imputation

A missing indicator is an extra binary column that records whether a value was originally missing. This allows the model to learn that "missingness" itself may be useful information. Random sample imputation fills missing values with values randomly chosen from the observed ones, which preserves the original distribution better than using a single mean.

7.5 KNN Imputer

K-Nearest Neighbors Imputer finds the rows most similar to the one with missing data (based on the other features) and averages their values. It is a multivariate method because it uses relationships between several columns.

7.6 MICE (Multivariate Imputation by Chained Equations)

MICE treats each column with missing values as a function of the other columns and models it iteratively. It repeatedly fills and refines the missing values until they stabilise. It is one of the most accurate imputation methods when columns are correlated.

8. Outliers

An outlier is a value that is unusually far from the other values. It may be a genuine rare case, a typo or a measurement error. Outliers can distort statistics and mislead some ML models.

8.1 Z-score Method

The Z-score measures how many standard deviations a value is from the mean. Values with $|Z| > 3$ are usually treated as outliers. This method assumes the data is roughly normal.

8.2 IQR Method

The Interquartile Range (IQR) is $Q3 - Q1$, where $Q1$ is the 25th percentile and $Q3$ is the 75th. Values below $Q1 - 1.5 \cdot IQR$ or above $Q3 + 1.5 \cdot IQR$ are considered outliers. It is robust and does not assume normality.

8.3 Percentile Method

Values outside chosen percentiles (for example the 1st and 99th) are treated as outliers. It is simple and robust to extreme values.

8.4 Winsorization

Instead of deleting outliers, Winsorization caps them at a lower and upper percentile (for example 1% and 99%). This keeps all rows while reducing the influence of extreme values, so no information is lost.

9. Variable Types and Date Handling

Datasets contain mixed variable types:

- Numerical variables: numbers on which arithmetic makes sense (age, income, credit score).
- Categorical variables: groups or labels (gender, region, loan purpose).
- Ordinal variables: categories with a natural order (education level).
- Datetime variables: dates and times (join date).

Datetime columns are not directly usable by models, so we extract useful parts such as year, month, day and weekday. We can also calculate tenure (how long a customer has been with the bank) by subtracting the join date from today.

10. Encoding Categorical Variables

Machine learning models work with numbers, so text categories must be converted.

10.1 Ordinal Encoding

Ordinal encoding assigns numbers while preserving order. For example: Primary = 0, Secondary = 1, Graduate = 2, Post-Graduate = 3. It is used when categories have a meaningful rank.

10.2 Label Encoding

Label encoding assigns a different integer to each category (0, 1, 2, ...). It is fine for binary or low-cardinality columns but can mislead models when used on nominal categories because it implies an order that does not exist.

10.3 One-Hot Encoding

One-hot encoding creates a separate binary column for each category. For example, region becomes region_North, region_South, region_East, region_West. It is used when categories have no natural order. We drop one column (drop_first) to avoid the dummy-variable trap, where columns become perfectly correlated.

11. Encoding Numerical Features

11.1 Binning (Discretization)

Binning groups continuous numeric values into a small number of intervals. For example income can be grouped as Low, Medium, High. It reduces noise and can reveal non-linear patterns.

11.2 Binarization

Binarization converts a numeric variable into a 0/1 flag using a threshold, for example good_credit_flag = 1 if credit_score > 700 else 0.

11.3 Quantile Binning

Quantile binning divides the data into bins so that each bin contains roughly the same number of observations. It is useful when data is skewed.

11.4 K-Means Binning

K-Means is an unsupervised clustering algorithm. It groups similar numeric values into K clusters. Each cluster then becomes a bin. It discovers natural groups in the data rather than using fixed cut-offs.

12. Feature Scaling

Feature scaling brings numeric columns to similar ranges. Without scaling, columns with large values (like income in lakhs) can dominate distance-based and gradient-based models.

- Standardization (Z-score): transforms values so the mean is 0 and standard deviation is 1. It is the default choice for many models.
- Min-Max Scaling (Normalization): rescales values to a fixed range, usually [0, 1]. Useful when the model needs bounded inputs.
- MaxAbs Scaling: scales by the maximum absolute value. It keeps the data centered around zero and is good for sparse data.
- Robust Scaling: uses the median and IQR instead of mean and standard deviation, so it is less affected by outliers.

Scaling does not change the shape of the distribution; it only changes its range.

13. Feature Construction

Feature construction means creating new columns from existing ones using domain knowledge. Well-designed features often improve model performance more than changing the algorithm.

- Debt-to-Income ratio = $\text{loan_amount} / \text{annual_income}$. A high ratio suggests higher risk.
- Spending-to-Income ratio = $\text{spending_ratio} / 100$. Shows what fraction of income is spent.
- Average monthly transactions = $\text{transaction_count} / 6$.
- Loan per age = $\text{loan_amount} / \text{age}$. Shows debt relative to the customer's life stage.
- Region default rate: an external feature from the dummy API showing historical default rates per region.

14. Feature Transformations

Some models (linear regression, logistic regression, neural networks) work better when numeric features are approximately normally distributed. Skewed columns can be transformed.

14.1 Log, Square Root and Reciprocal

A log transform compresses large values and reduces right-skewness. Square root is milder, while reciprocal ($1/x$) changes the shape more strongly. A small constant is often added to avoid $\log(0)$ or division by zero.

14.2 Box-Cox and Yeo-Johnson (Power Transforms)

Box-Cox automatically finds the best power transformation to make a positive column look more Gaussian. It requires strictly positive values. Yeo-Johnson is similar but also works with zero and negative values. Both reduce skewness and help linear and distance-based models.

15. Column Transformer and Pipelines

A ColumnTransformer applies different preprocessing steps to different groups of columns in a single object. For example:

- Numeric columns: median imputation followed by standard scaling.
- Categorical columns: most-frequent imputation followed by one-hot encoding.

A Pipeline chains these steps together. The result is clean, reproducible and easy to deploy: new data can be transformed with one call. Pipelines also prevent data leakage because all fitting happens only on the training data.

16. Final Dataset and ML Readiness

After all the preprocessing steps the final dataset should have:

- No missing values.
- No unhandled outliers.
- All columns numeric and correctly encoded.
- Similar-scaled numeric columns.
- New meaningful features.
- A clearly separated target column (default_flag).

Such a dataset is said to be "ML-ready". It can directly be used to train classification models such as Logistic Regression, Decision Trees, Random Forest, Gradient Boosting or XGBoost to predict whether a customer will default on a loan.

17. Summary

Data preprocessing and feature engineering are the most time-consuming but also the most important parts of a machine learning project. Clean, well-engineered data helps even simple models perform well, while poor data will limit even the most advanced algorithms. This project demonstrates the complete workflow — acquiring data from multiple sources, understanding and cleaning it, handling missing values and outliers, encoding and scaling features, constructing new variables and packaging everything into a reusable pipeline — so the final dataset is ready for machine learning.